

SESSION 13

CRUD Using AJAX

Topics Covered: AJAX introduction, Fetch API, Sending JSON request, Handling JSON responses

Activities: Implement delete doctor using AJAX, Show success message dynamically

Task 1: Create simple HTML page with list of songs (Spotify playlist style) and add 'Delete' button next to each song; use fetch() to send AJAX DELETE request.

Explanation & Approach: *AJAX Fetch API allows asynchronous HTTP requests without triggering full web page reloads.*

```
<!-- song_list.html -->
<ul id="song-list">
  <li id="song-1">
    Bohemian Rhapsody <button onclick="deleteSong(1)" class="btn btn-sm btn-
danger">Delete</button>
  </li>
</ul>

<script>
function deleteSong(songId) {
  fetch(`/api/songs/${songId}/`, {
    method: 'DELETE',
    headers: {
      'X-CSRFToken': '{{ csrf_token }}'
    }
  })
  .then(response => {
    if(response.ok) {
      document.getElementById(`song-${songId}`).remove();
    }
  });
}
</script>
```

Task 2: Build mini Flipkart product list with Python and JS, where clicking 'Remove from Wishlist' sends AJAX request and removes item without reload.

Explanation & Approach: *Combines Django JsonResponse endpoint with DOM node removal.*

```
# views.py
from django.http import JsonResponse

def remove_wishlist(request, product_id):
    if request.method == 'DELETE':
        # Remove item from wishlist model/session
        return JsonResponse({'status': 'success', 'message': 'Product removed from
wishlist'})
    return JsonResponse({'status': 'error'}, status=400)
```

Task 3: Implement feature for BookMyShow movie booking page where deleting movie from 'Watch Later' uses AJAX (fetch API) to send JSON and show dynamic success message.

Explanation & Approach: *Returns JSON status and renders dynamic dismissible alert banner in DOM.*

```
// JavaScript fetch request with JSON response handling
function removeFromWatchLater(movieId) {
  fetch(`/watch-later/delete/${movieId}/`, {
    method: 'POST',
    headers: {
      'Content-Type': 'application/json',
      'X-CSRFToken': getCookie('csrftoken')
    }
  })
  .then(res => res.json())
  .then(data => {
    if (data.success) {
      document.getElementById(`movie-card-${movieId}`).remove();
      showNotification(data.message); // Displays dynamic success message banner
    }
  });
}
```

Task 4: Debug scenario: function sends fetch() DELETE request to remove playlist from music app, but playlist is not disappearing from page. Identify and fix issue.

Explanation & Approach: *Fixes missing DOM removal logic inside the fetch promise resolution block.*

```
// BUGGY CODE:
// fetch('/playlist/delete/1/', {method: 'DELETE'}).then(res => console.log('Deleted'));
// Cause: Missing DOM manipulation logic in .then() block!

// FIXED CODE:
function deletePlaylist(playlistId) {
  fetch(`/playlist/delete/${playlistId}/`, {
    method: 'DELETE',
    headers: { 'X-CSRFToken': getCookie('csrftoken') }
  })
  .then(response => response.json())
  .then(data => {
    if (data.status === 'deleted') {
      const element = document.getElementById(`playlist-item-${playlistId}`);
      if (element) {
        element.remove(); // FIX: Explicitly remove DOM element on success
      }
    }
  })
  .catch(error => console.error('Error deleting playlist:', error));
}
```